Elevation resonances of the LUCY mount as fork modes driven by the motor stator: a data-driven lumped model with two stator modal masses, and a flexible azimuth load

Generated by `build_fork_modes_report.py`

`sim/gimbal_multidof/report_fork_modes`

October 29, 2026

Purpose. Frequency responses taken with the motion of the fork constrained showed that the resonances of the elevation axis below 150 Hz are vibrations of the fork, which also explains why resonances at the same frequencies appear on the azimuth axis. This report describes how a data-driven model of those resonances was built and identified:

- on the elevation axis, the lumped model H7 is extended with *modal masses attached to the motor stator*. The reaction of the motor torque excites them, and they move the encoder read head, which is mounted on the same structure;
- on the azimuth axis, the load of H7 is split into a fork base and a fork top joined by a spring, so that the load is no longer rigid.

The step documented here is the reduced elevation model with **two** stator masses. Of the four masses of the previous iteration (8.8, 53.4, 57.5, 100.2 Hz), the ones at 8.8 and 53.4 Hz were removed and the model was refitted. All the results refer to the fit band **1–150 Hz**.

Main results.

- **Elevation.** Two stator masses, at 57.7 Hz and 100.1 Hz, with residues of opposite sign.
  - The fit (NRMSE on the complex response, 1–150 Hz) on the three nominal measurements goes from 38, −13 and −12 % with H7 to 57, 69 and 52 %.
  - The rms errors are 1.7–2.2 dB in magnitude and 7.6–17.9° in phase.
  - All 16 starts of the least squares end in the same minimum.
- **Non-collocated mode.** In the mode at 57.7 Hz the encoder head moves against the stator (negative residue).
  - With the mode at 100.1 Hz, this produces a quartet of zeros, 74.6 Hz ($\zeta = -0.26$), 81.8 Hz ($\zeta = +0.35$): one pair is in the right half-plane.
  - This pair explains the extra phase drop of about 360° measured between 45 and 120 Hz.
- **Azimuth.** With the flexible load the fit goes from −21 % (H7) to 70 % (2.1 dB, 8.6° rms).
  - The fork mode at 55.8 Hz has the motor and the fork base moving against the fork top and the telescope.
  - It lies at the frequency of the elevation stator mass at 57.7 Hz: the same fork vibration is seen by both axes.
- **Cost of the reduction.** Removing the masses at 8.8 and 53.4 Hz loses the antiresonance–resonance pair at 8–9 Hz.
  - The fit of the 1–600 Hz measurement between 5 and 100 Hz drops from 66 to 32 %.
  - Above 30 Hz the two-mass model is as good as the four-mass one.

Note. The H7 configuration contains placeholder data: the inertias of the encoder and motor nodes, the materials, and the elevation during the tests (taken as $\theta_0 = 0$). The figures therefore carry the label "result from placeholder parameters". The inertias are kept fixed at the CAD/H7 values in all the fits.

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1 Data

Four frequency responses measured with Kollmorgen WorkBench on 28/10/2026 (folder `LucyIdentification/Poli_Polar_identification`), all taken 12 mm from the end stop (Table 1). The measured quantity is the WorkBench trace "Plant":

- the encoder velocity over the commanded current, in (rad/s)/A;
- measured with a stepped-sine current excitation, an FFT of 4096 points and 50 % overlap;
- smoothed by WorkBench by 1 % in frequency.

$$G_{\mathrm{mis}}(j\omega) = \frac{\Omega_{\mathrm{enc}}(j\omega)}{I(j\omega)}. \tag{1.1}$$

Use of the data.

- **Band.** Only the points between 1 and 150 Hz are used: 748 points on the elevation axis and 170 on the azimuth axis.
- **Elevation.** The three elevation measurements are fitted together with one set of parameters. Two of them overlap the first between 30 and 150 Hz: one repeats it on a finer grid, the other was taken with all 16 screws and 4 A.
- **Coherence.** The exported files do not contain the coherence, so the points cannot be weighted by their quality.

Table 1: Measurements used. Grid: median step between points. Points: number of points in the fit band 1–150 Hz.

axis	file	range [Hz]	grid [%]	current [A]	points
elevation	`BodeMeasurement_Polar_14vit_600_20261028_1perc.csv`	1–595	1	3	504
elevation	`BodeMeasurement_Polar_14vit_30_150_20261028.csv`	30–149	1	3	162
elevation	`BodeMeasurement_Polar_14vit_78_20261028_2perc_12mm_tightenedlaser_newsupport_4A_80degs_16screws.csv`	30–298	2	4	82
azimuth	`BodeMeasurement_Azimuth_14vit_600_20261028.csv`	1–593	3	3.5	170

2 Model

2.1 Measured plant

Both axes are written as linear lumped models,

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{e}\,\tau, \qquad \theta_{\text{enc}} = \mathbf{c}^T\mathbf{x}, \qquad G_\theta(s) = \frac{\theta_{\text{enc}}}{\tau} = \mathbf{c}^T\big(\mathbf{M}s^2 + \mathbf{C}s + \mathbf{K}\big)^{-1}\mathbf{e}, \tag{2.1}$$

where τ is the motor torque, $\mathbf{e}$ says where it acts and $\mathbf{c}$ what the encoder reads. $\mathbf{K}$ may be complex: the bearing constraint is hysteretic (Section 2.2). The plant compared with the measurement (Figure 1) adds:

- the derivative from angle to velocity;
- the torque constant $K_t = 0.258\,\text{N m/A}$;
- the current loop, a first-order low-pass at $\omega_i = 2\pi \cdot 1.5\,\text{kHz}$;
- an equivalent pure delay τ_d, which takes up the sampling and the dynamics above the band.

$$G(s) = s\,K_t\,G_\theta(s)\,\frac{\omega_i}{s+\omega_i}\,e^{-s\tau_d}. \tag{2.2}$$

The forward anti-resonance filters active during the tests are not part of the WorkBench "Plant" trace, so they are not applied.

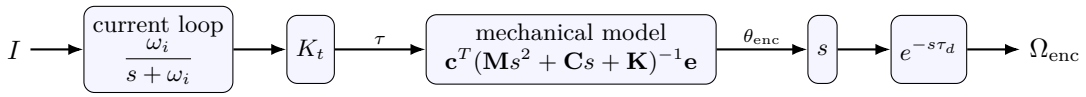


Figure 1: Plant measured by WorkBench ("Plant" trace) as modelled in (2.2). The mechanical block is the elevation model of Figure 2 or the azimuth model of Figure 3.

2.2 Elevation: H7 with modal masses on the motor stator

Rotor chain (H7). The elevation block of H7 has five nodes: encoder disc e, hub centre h, motor rotor m, rear end of the shaft (rear), telescope p (Figure 2).

- Their inertias are fixed at the CAD/H7 values. The rigid inertia of the axis is $J = 0.900\,\text{kg m}^2$.
- The shaft stiffnesses and dampers $k_{\theta1}, c_{\theta1}$ (e–h), $k_{\theta2}, c_{\theta2}$ (h–m) and $k_{\theta3}, c_{\theta3}$ (m–rear) are also those of H7.
- The joint between the telescope and the hub, k_b and c_b, is fitted.

Stator masses. Each mode of the fork excited by the stator is a modal oscillator with coordinate q_r:

- it has modal inertia J_r, modal stiffness k_r and modal damping c_r, all positive, with respect to the ground;
- the stator reaction $-\tau$ acts on all of them with unit participation;
- the rotation of the stator structure is the modal superposition $\varphi_s = \sum_r q_r$.

Bearings. The polar bearings are mounted on the same structure. Their pre-sliding friction is a hysteretic spring $k_g^* = k_g(1 + j\eta_g)$ between the hub and the stator structure, not the ground.

Encoder. The disc turns with θ_e, while the read head is carried by the structure and moves by $\sum_r \kappa_r q_r$. The encoder therefore reads

$$\theta_{\text{enc}} = \theta_e - \sum_{r=1}^{N} \kappa_r q_r. \tag{2.3}$$

κ_r is the rotation of the head per unit of modal coordinate, a kinematic factor and not a stiffness. A negative κ_r means that in mode r the head turns against the point where the stator reaction is applied: the motor and the sensor are not collocated for that mode.

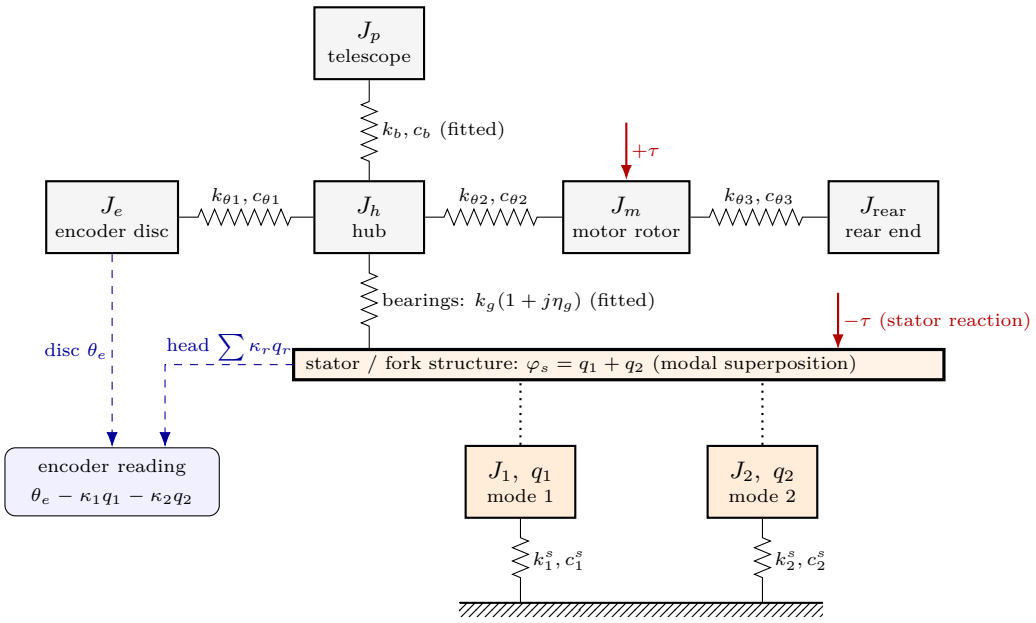


Figure 2: Elevation model, drawn as its translational analogue (each spring is a spring and a viscous damper in parallel). Grey: rotor chain of H7. Orange: stator structure and its modal oscillators, each grounded by $k_r^s = J_r\omega_r^2$ and $c_r^s = 2\zeta_r J_r \omega_r$. The motor torque acts on the rotor ($+\tau$) and its reaction on the stator ($-\tau$). The bearings join the hub to the stator structure. The encoder reads the disc minus the motion of the head carried by the structure.

Equations. With $\mathbf{x} = [\theta_e, \theta_h, \theta_m, \theta_{\text{rear}}, \theta_p, q_1, \dots, q_N]^T$, the modal oscillators obey

$$J_r\ddot{q}_r + c_r^s\dot{q}_r + k_r^s q_r = -\tau + k_g^*\Big(\theta_h - \sum_{s=1}^{N} q_s\Big), \qquad k_r^s = J_r\omega_r^2, \quad c_r^s = 2\zeta_r J_r\omega_r, \tag{2.4}$$

and the hub receives the opposite bearing torque $-k_g^*(\theta_h - \sum_s q_s)$. In the form (2.1),

$$\mathbf{M} = \begin{bmatrix}\mathbf{M}_0 & \\ & \operatorname{diag}(J_r)\end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix}\mathbf{C}_0(c_b) & \\ & \operatorname{diag}(c_r^s)\end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix}\mathbf{K}_0(k_b) & \\ & \operatorname{diag}(k_r^s)\end{bmatrix} + k_g^*\,\mathbf{v}\mathbf{v}^T, \tag{2.5}$$

$$\mathbf{v} = \mathbf{e}_h - \sum_r \mathbf{e}_{q_r}, \qquad \mathbf{e} = \mathbf{e}_m - \sum_r \mathbf{e}_{q_r}, \qquad \mathbf{c} = \mathbf{e}_e - \sum_r \kappa_r\,\mathbf{e}_{q_r}, \tag{2.6}$$

where $\mathbf{M}_0, \mathbf{C}_0, \mathbf{K}_0$ are the elevation block of H7 with its k_b, c_b replaced by the fitted ones, and $\mathbf{e}_j$ is the unit vector of coordinate j.

What the measurement can determine. Neglecting the bearing coupling ($k_g \ll k_r^s$: here 117 against $\geq 10^5$ N m/rad), mode r adds to G_θ the term

$$-\kappa_r\frac{q_r}{\tau} = \frac{\kappa_r/J_r}{s^2 + 2\zeta_r\omega_r s + \omega_r^2} = \frac{a_r}{s^2 + 2\zeta_r\omega_r s + \omega_r^2}, \qquad a_r = \frac{\kappa_r}{J_r}. \tag{2.7}$$

The measurement fixes the modal residue a_r, not J_r and κ_r separately. The fit therefore estimates (f_r, ζ_r, a_r) for each mass, and the scale is set by the convention $|\kappa_r| = 1$, i.e. $J_r = 1/|a_r|$ and $\kappa_r = \operatorname{sign}(a_r)$. The J_r of the tables are modal inertias under that convention, not physical inertias of parts.

2.3 Sign of the residue and non-minimum-phase zeros

Above a few hertz the telescope moves with the hub (Section 4), and the rotor behaves as the rigid inertia J. With the two masses, undamped,

$$G_\theta(s) \simeq \frac{1}{Js^2} + \frac{a_1}{s^2 + \omega_1^2} + \frac{a_2}{s^2 + \omega_2^2} = \frac{Au^2 + Bu + C}{s^2(s^2 + \omega_1^2)(s^2 + \omega_2^2)}, \qquad u = s^2, \tag{2.8}$$

$$A = \frac{1}{J} + a_1 + a_2, \qquad B = \frac{\omega_1^2 + \omega_2^2}{J} + a_1\omega_2^2 + a_2\omega_1^2, \qquad C = \frac{\omega_1^2\omega_2^2}{J}. \tag{2.9}$$

- **Positive residues** (collocated actuator and sensor). The zeros interlace with the poles on the imaginary axis (u real and negative): every resonance is preceded by an antiresonance, and the phase comes back after each pair.

- **The fitted values** ($J = 0.900\,\mathrm{kg\,m^2}$, $a_1 = -0.695$, $a_2 = +0.626\,(\mathrm{kg\,m^2})^{-1}$) give $A = 1.043$, $B = 3.94\times10^5$ and $C = 5.79\times10^{10}$, so $B^2 - 4AC = -8.7\times10^{10} < 0$.
  - The two roots u are complex, and the four zeros $s = \pm\sqrt{u}$ form a quartet $\pm\sigma \pm j\omega$: one pair lies in the right half-plane.
  - In this approximation the quartet has $|s|/2\pi = 77.3\,\mathrm{Hz}$ and $\zeta = \pm0.32$.
  - The full model gives 74.6 Hz ($\zeta = -0.26$), 81.8 Hz ($\zeta = +0.35$) (Figure 7).
- **Effect on the phase.** The quartet contributes $+180 - 180 = 0°$ instead of the $+360°$ of two collocated antiresonances. Across the two modes the phase therefore falls by $3\times180°$, against $180°$ for a collocated pattern.

This is the drop measured between 45 and 120 Hz, which a model with positive residues only cannot reproduce: a fit restricted to $\kappa_r > 0$ ended in a poor local minimum.

2.4 Azimuth: flexible load

In H7 the azimuth load (fork, hub and ring, $J_{\mathrm{load}} = 1.305\,\mathrm{kg\,m^2}$ from CAD) is one rigid node L, with the telescope P attached to it. Here it is split into a fork base L and a fork top A (Figure 3):

$$J_L = (1-\alpha)\,J_{\mathrm{load}}, \qquad J_A = \alpha\,J_{\mathrm{load}}, \tag{2.10}$$

- the two parts are joined by k_{LA}, c_{LA}, and the total inertia stays the CAD value;
- the telescope hangs from the fork top through k_3, c_3;
- the bearing constraint $k_g(1 + j\eta_g)$ grounds the fork base;
- the motor shaft k_2, c_2 is fitted, while the encoder shaft $k_1 = 9.55\times10^3\,\mathrm{N\,m/rad}$ and the inertias J_e, J_m, J_P are those of H7.

With $\mathbf{x} = [\psi_e, \psi_m, \psi_L, \psi_A, \psi_P]^T$, the model (2.1) has $\mathbf{M} = \mathrm{diag}(J_e, J_m, J_L, J_A, J_P)$ and a chain $\mathbf{K}$, $\mathbf{C}$ of the four links, with k_g^* added at (L, L); $\mathbf{e} = \mathbf{e}_m$ and $\mathbf{c} = \mathbf{e}_e$.

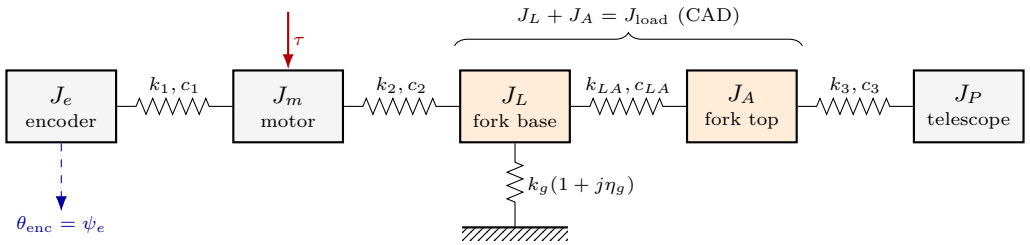


Figure 3: Azimuth model with the flexible load, drawn as its translational analogue (each spring is a spring and a viscous damper in parallel). Orange: the H7 load node split into fork base and fork top.

3 Identification

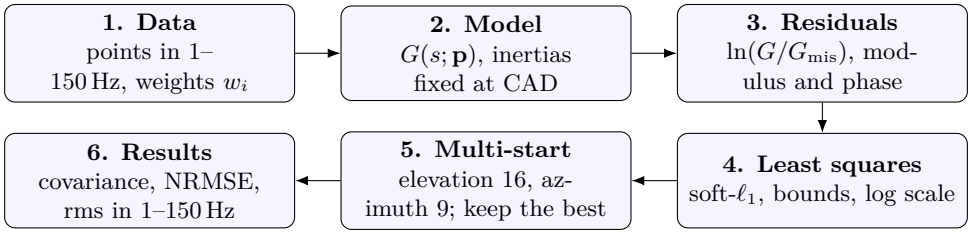


Figure 4: Flow of the identification, run separately for the two axes by `analisi_modi_forcella.py` (model in `src/gimbal_mdof/modi_forcella.py`).

3.1 Error and cost

Error. At each measured frequency f_i the error is the complex logarithm of the ratio of model to measurement:

$$\varepsilon_i = \ln\frac{G(j\omega_i;\mathbf{p})}{G_{\mathrm{mis}}(j\omega_i)} = \underbrace{\ln\left|\frac{G}{G_{\mathrm{mis}}}\right|}_{\text{magnitude, Np}} + j\,\underbrace{\left(\angle G - \angle G_{\mathrm{mis}}\right)}_{\text{phase, rad}}. \tag{3.1}$$

- Magnitude and phase weigh alike: $1\,\mathrm{Np} = 8.7\,\mathrm{dB}$ counts as $1\,\mathrm{rad} = 57.3°$.
- The phase difference is taken on the ratio, so it does not depend on the unwrapping.
- Antiresonances (small $|G|$) count as much as resonances, unlike an error on G itself.

Weights. The grids differ (1 to 3 % steps) and the three elevation measurements overlap. Each point is weighted by the width of the frequency interval it represents:

$$w_i = \sqrt{\frac{\Delta \ln f_i}{0.01}}, \tag{3.2}$$

so that the cost is an integral over $\ln f$, and a finer grid does not weigh more than a coarser one over the same band. The residual vector is $\mathbf{r} = [\, w_i \operatorname{Re}\varepsilon_i,\ w_i \operatorname{Im}\varepsilon_i \,]$ over all the points of all the measurements of the axis.

Cost. A robust soft-ℓ_1 loss limits the weight of the points the model cannot follow (the 8–9 Hz pair, the small features at 48–50 and 84 Hz):

$$F(\mathbf{p}) = \frac{1}{2}\sum_j C^2 \rho\left(\frac{r_j^2}{C^2}\right), \qquad \rho(z) = 2\big(\sqrt{1+z} - 1\big), \qquad C = 0.1, \tag{3.3}$$

quadratic for $|r_j| \ll C$ ($0.1\,\mathrm{Np} \approx 0.9\,\mathrm{dB}$, $0.1\,\mathrm{rad} \approx 5.7°$) and linear above.

3.2 Parameters, bounds and starts

- **Scale.** The positive parameters are fitted on their logarithm (relative steps, positivity). The residues a_r, which carry a sign, are fitted on their value.
- **Elevation.**
  - Fitted: k_b, c_b, k_g, η_g, τ_d, and (f_r, ζ_r, a_r) of each mass.
  - Bounds on f_r: $\pm 10\,\%$ of the start (58.5 and 99.2 Hz, the measured resonances). Bounds on a_r: $[-50, 50]$.
  - 16 starts: $k_b \in \{1.16 \times 10^5, 10^6\}\,\mathrm{N\,m/rad}$, both signs of each residue, and $|a_r| \in \{0.3, 1.5\}$, below and above $1/J = 1.111$.
- **Azimuth.**
  - Fitted: α, k_{LA}, c_{LA}, k_3, c_3, k_2, c_2, k_g, η_g, τ_d.
  - 9 starts: $\alpha \in \{0.2, 0.4, 0.7\}$ combined with three assignments of the measured 49 and 56 Hz resonances to the base–top link and to the telescope: (49, 56), (56, 49), and (56 Hz, telescope stiff at 120 Hz).
- **Solver.** Trust-region reflective least squares (`scipy.optimize.least_squares`, Jacobian scaling), at most 800 evaluations per start. The start with the lowest cost is kept.

The bounds and the starts of every parameter are in Tables 2 and 4.

3.3 Uncertainty and goodness measures

Uncertainty. From the Jacobian $\mathbf{J}$ of the residuals at the optimum,

$$\operatorname{cov}(\mathbf{x}) = s^2\,(\mathbf{J}^T\mathbf{J})^{+}, \qquad s^2 = \frac{2F}{m-n}, \tag{3.4}$$

with m residuals and n parameters. For a parameter fitted on its logarithm, the standard deviation of $\ln p$ is its relative standard deviation. A value above 100 % means that the data do not determine the parameter.

Fit. The NRMSE fit of the System Identification Toolbox, on the complex response at the measured points of a band:

$$\text{fit} = 100\left(1 - \frac{\|G_{\mathrm{mis}} - G\|}{\|G_{\mathrm{mis}} - \overline{G_{\mathrm{mis}}}\|}\right)\ \%. \tag{3.5}$$

- It is reported over the whole fit band (1–150 Hz) and in the sub-bands 1–5, 5–100 and 100–150 Hz.
- On the complex response it is dominated by the points of large magnitude (resonances, low frequency).
- The rms of $20\log_{10}|G/G_{\mathrm{mis}}|$ (dB) and of the phase error (deg) weigh all the points alike.

4 Results: elevation

Convergence. All 16 starts end at the same cost, 18.67. The minimum does not depend on the signs or moduli of the starting residues. The best start places a negative residue on the mass at 57.7 Hz and a positive one on the mass at 100.1 Hz.

Table 2: Elevation: fitted parameters. H7: value in the configuration. Start: values of the 16 starts. Std: relative standard deviation from the Jacobian.

parameter	unit	estimate	H7	start	bounds	std	at bound
k_b	N m/rad	1×10^4	1.16×10^5	1.16×10^5, 1×10^6	$[1 \times 10^4, 1 \times 10^9]$	$> 10^3$ %	yes
c_b	N m s/rad	2260	1.254	1.25	$[0.001, 1 \times 10^4]$	12.2 %	
k_g	N m/rad	116.7	164	164	$[1, 1 \times 10^5]$	8.5 %	
η_g	–	3.201	3.11	3.11	[0.001, 30]	8.3 %	
τ_d	s	0.0005945	–	0.0002	$[1 \times 10^{-6}, 0.001]$	4.4 %	
f_1	Hz	57.74	–	58.5	[52.6, 64.4]	0.2 %	
ζ_1	–	0.03773	–	0.01	[0.001, 0.5]	6.0 %	
a_1	$1/(\mathrm{kg\,m^2})$	-0.6948	–	± 0.3, ± 1.5	[-50, 50]	2.1 %	
f_2	Hz	100.1	–	99.2	[89.3, 109]	0.2 %	
ζ_2	–	0.03021	–	0.01	[0.001, 0.5]	5.0 %	
a_2	$1/(\mathrm{kg\,m^2})$	0.6264	–	± 0.3, ± 1.5	[-50, 50]	2.2 %	

Table 3: Elevation: stator masses. $J_r = 1/|a_r|$ and $\kappa_r = \mathrm{sign}(a_r)$ under the convention $|\kappa_r| = 1$; $k_r^s = J_r \omega_r^2$, $c_r^s = 2\zeta_r J_r \omega_r$.

mass	f_r [Hz]	ζ_r	a_r [$1/(\mathrm{kg\,m^2})$]	J_r [$\mathrm{kg\,m^2}$]	κ_r	k_r^s [N m/rad]	c_r^s [N m s/rad]
1	57.74	0.0377	-0.695	1.439	-1	1.89×10^5	39.4
2	100.12	0.0302	+0.626	1.596	+1	6.32×10^5	60.7

Stator masses.

- **Mass 1** ($f_1 = 57.7$ Hz, $\zeta_1 = 0.038$, $a_1 = -0.695$). It takes up the measured pair of resonances at 56.3 and 58.5 Hz as a single mode: the model peak (57.6 Hz) falls between the two measured ones.
  - Its negative residue makes it non-collocated (Section 2.3).
  - $a_1 J = -0.63$: its residue is 63 % of the rotor one, $1/J$, with the opposite sign, which is large enough to produce the right-half-plane zeros.
- **Mass 2** ($f_2 = 100.1$ Hz, $\zeta_2 = 0.030$, $a_2 = +0.626$) is collocated.
- Frequencies, damping ratios and residues are determined within 6 %.

Telescope joint. k_b goes to its lower bound and is not determined, while $c_b = 2260$ N m s/rad. The joint acts as a damper that couples the telescope to the hub up to $c_b/J_p \approx 2\pi \cdot 400$ Hz. Below 150 Hz the data only require that the telescope moves with the hub, and the antiresonance of H7 at 58 Hz (from $k_b = 1.16 \times 10^5$ N m/rad) is not seen. The four-mass fit reached the same conclusion with a different pair ($k_b = 8 \times 10^5$ N m/rad, $c_b = 717$ N m s/rad).

Bearings and delay.

- $k_g = 117$ N m/rad, $\eta_g = 3.20$, close to the ground constraint of the configuration (164 N m/rad, 3.11). They set the low-frequency rigid-body pole at 1.8 Hz.
- The same model with the bearings to the ground instead of to the stator gives the same cost (18.66 against 18.67). With two masses the data cannot tell where the bearings are attached.
- The equivalent delay is $\tau_d = 0.59$ ms.

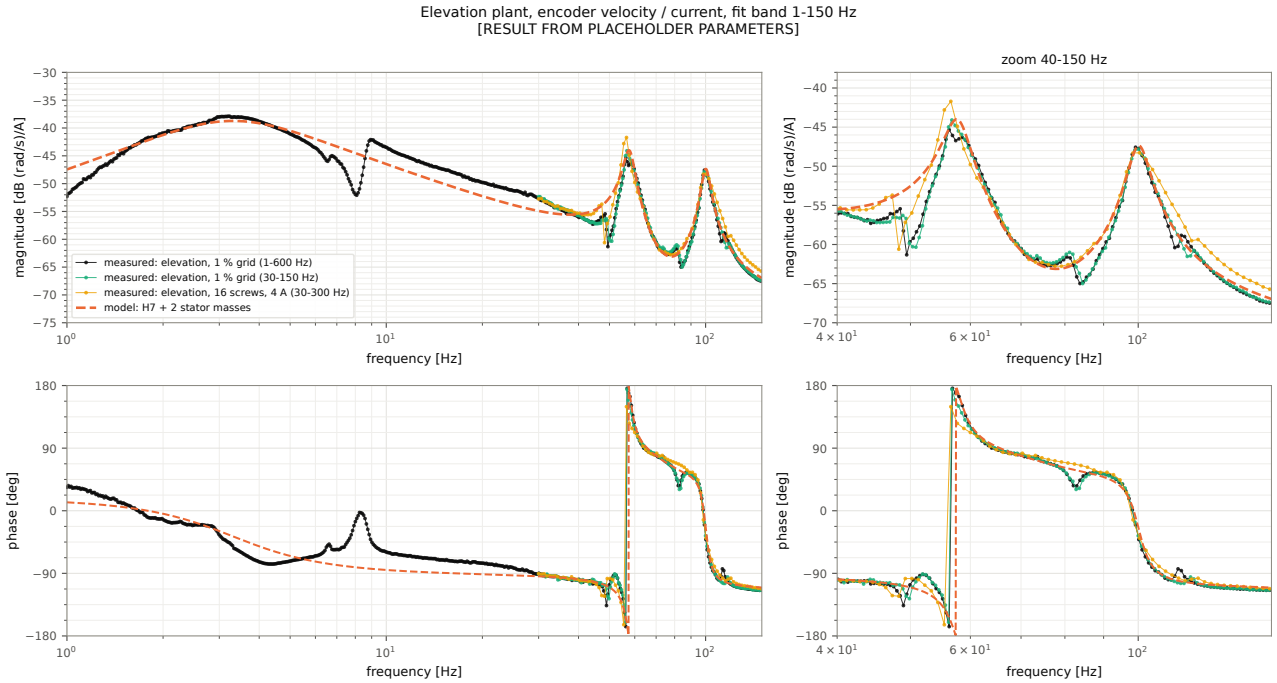


Figure 5: Elevation plant [dB (rad/s)/A] and phase [deg] in the fit band: the three measurements and the model with two stator masses. Right: zoom 40–150 Hz.

Decomposition of the encoder reading (Figure 6).

- The rotor term θ_e carries the whole response up to about 40 Hz.
- Mass 1 dominates around 57 Hz.
- Between 60 and 90 Hz, the term of mass 1 (phase near +90°, Figure 6) and the term of mass 2 partly cancel the rotor term. This is where the non-minimum-phase quartet lies.
- Mass 2 dominates around 100 Hz.

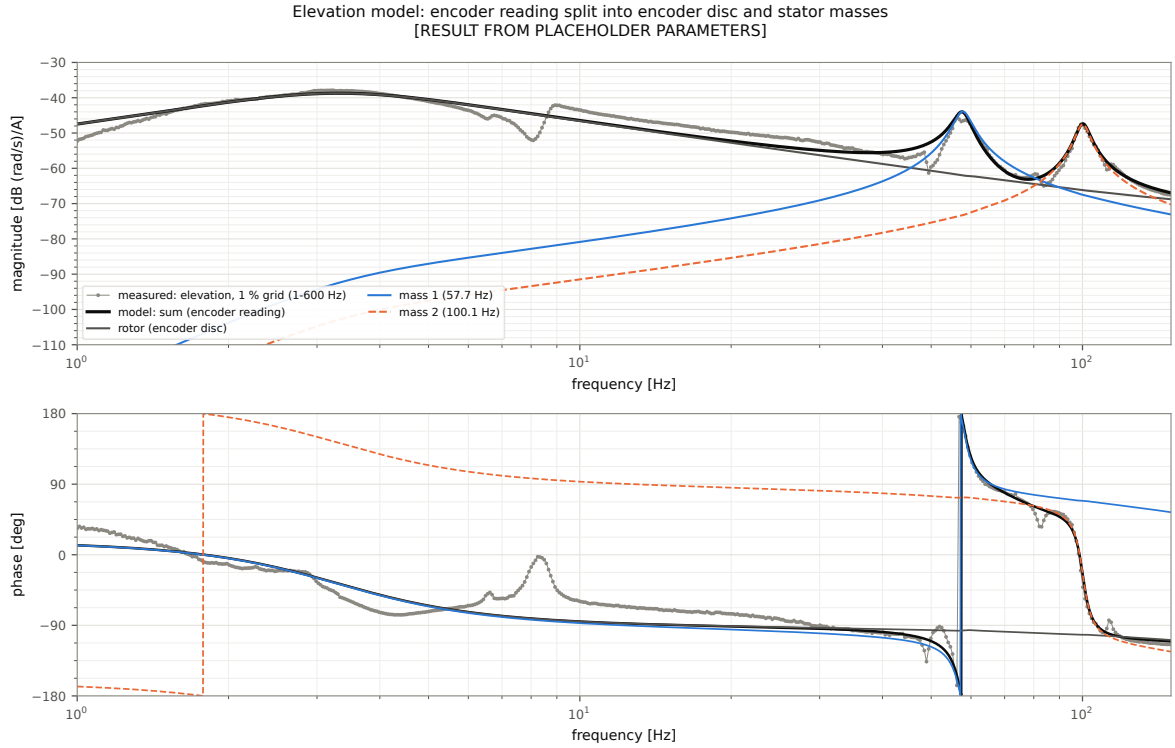


Figure 6: Elevation model: encoder reading (2.3) [dB (rad/s)/A, deg] split into the encoder disc term (rotor) and the head motion of each stator mass, $-\kappa_r q_r$, each with the plant factors of (2.2). Grey dots: 1–600 Hz measurement in the band.

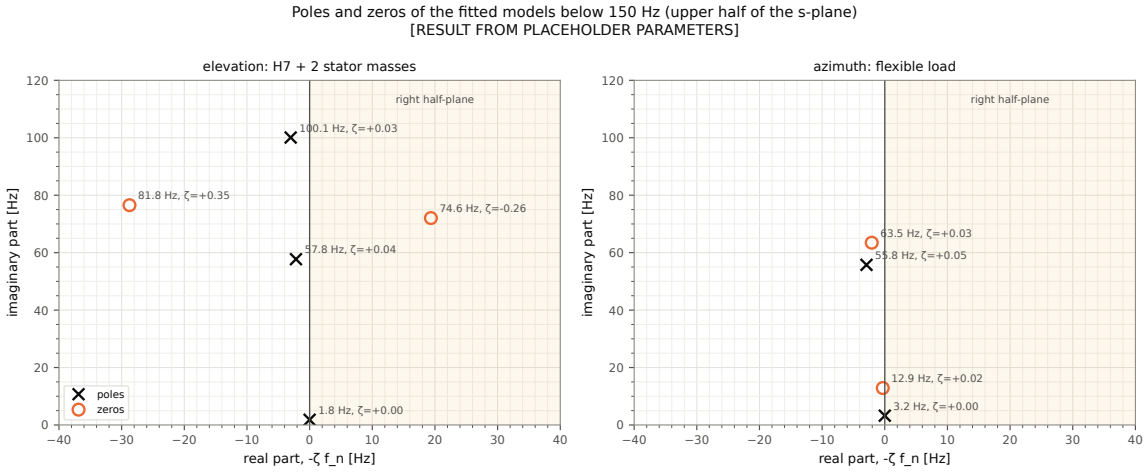


Figure 7: Poles (crosses) and zeros (circles) of G_θ below 150 Hz, upper half of the s-plane, in Hz, with the damping ratio. Shaded: right half-plane. Elevation: the zero at 74.6 Hz with $\zeta < 0$ and its mirror image at 81.8 Hz form the quartet of Section 2.3.

5 Results: azimuth

Table 4: Azimuth: fitted parameters. H7: value in the configuration. Start: values of the 9 starts. Std: relative standard deviation from the Jacobian.

parameter	unit	estimate	H7	start	bounds	std	at bound
α	–	0.659	–	0.2, 0.4, 0.7	[0.02, 0.95]	4.2 %	
k_{LA}	N m/rad	4.179×10^4	–	6 values, 2.5×10^4–1.1×10^5	$[1 \times 10^3, 1 \times 10^8]$	4.6 %	
c_{LA}	N m s/rad	0.001	–	1	$[0.001, 1 \times 10^3]$	$> 10^3$ %	yes
k_3	N m/rad	2.861×10^4	1.16×10^5	8.13×10^4, 1.06×10^5, 4.88×10^5	$[1 \times 10^3, 1 \times 10^8]$	21.9 %	
c_3	N m s/rad	278.3	4.901	4.9	$[0.001, 1 \times 10^3]$	62.8 %	
k_2	N m/rad	1.884×10^4	1.309×10^4	1.31×10^4	$[6.55 \times 10^3, 3.93 \times 10^4]$	3.2 %	
c_2	N m s/rad	0.001	0.5531	0.553	[0.001, 100]	$> 10^3$ %	yes
k_g	N m/rad	914.4	1418	1.42×10^3	$[10, 1 \times 10^5]$	3.2 %	
η_g	–	0.6449	0.32	0.32	[0.001, 10]	4.6 %	
τ_d	s	0.0004534	–	0.0001	$[1 \times 10^{-6}, 0.002]$	84.4 %	

Load split. The fork top takes $\alpha = 0.659$ of the CAD load:

- $J_A = 0.860\,\mathrm{kg\,m^2}$ for the fork top and $J_L = 0.445\,\mathrm{kg\,m^2}$ for the fork base;
- the base–top link has $k_{LA} = 4.18 \times 10^4\,\mathrm{N\,m/rad}$;
- the motor shaft comes out stiffer than in H7, $k_2 = 1.88 \times 10^4$ against $1.31 \times 10^4\,\mathrm{N\,m/rad}$.

Two dampers, c_{LA} and c_2, go to their lower bound and are not determined: the damping of the fork mode is supplied by c_3.

Modes (Table 5).

- The rigid-body mode on the bearing constraint is at 3.2 Hz.
- The fork mode is at 55.8 Hz, with $\zeta = 0.052$. In it the encoder, the motor and the fork base move together against the fork top and the telescope.
- This mode gives the measured antiresonance–resonance–antiresonance pattern: 12.3/55.7/64.6 Hz measured, 12.9/54.8/64.0 Hz in the model (Table 7).

Table 5: Azimuth: modes of the fitted model below 150 Hz (real stiffness of the constraint), shapes normalised on the largest node.

mode	f [Hz]	ζ	e	m	L	A	P
1	3.24	0.0005	+0.97	+0.97	+0.97	+0.99	+1.00
2	55.83	0.0523	+1.00	+1.00	+1.00	-0.28	-0.23

Telescope joint and second minimum. The best fit gives $k_3 = 2.86 \times 10^4\,\mathrm{N\,m/rad}$ with $c_3 = 278\,\mathrm{N\,m\,s/rad}$. This is an overdamped telescope, $\zeta \approx c_3/(2\sqrt{k_3 J_P}) \approx 0.9$, with a standard deviation of 22 % on k_3 and 63 % on c_3. The 9 starts end in three minima, with costs 6.11, 6.70, 6.77:

- The second (+10 %) keeps the telescope joint near its H7 value: $k_3 = 1.18 \times 10^5\,\mathrm{N\,m/rad}$, $c_3 = 22.6\,\mathrm{N\,m\,s/rad}$, $\alpha = 0.726$.
- It adds a lightly damped telescope mode at 86.8 Hz ($\zeta = 0.059$) that the measurement does not show clearly.
- The data in the band discriminate only weakly between the two descriptions of the telescope.

Bearings and delay. $k_g = 914\,\mathrm{N\,m/rad}$ and $\eta_g = 0.64$; the ground constraint of the configuration is $1418\,\mathrm{N\,m/rad}$ and 0.32. The delay $\tau_d = 0.45\,\mathrm{ms}$ is poorly determined (84 %).

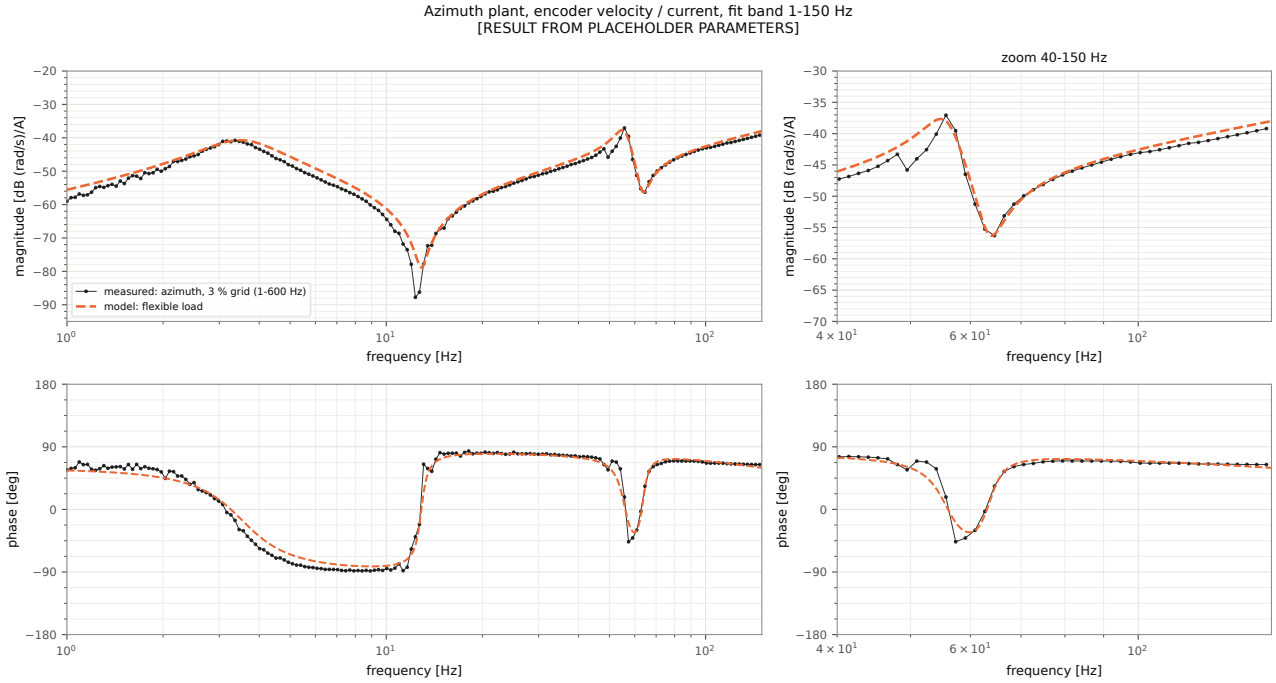


Figure 8: Azimuth plant [dB (rad/s)/A] and phase [deg] in the fit band: measurement and model with the flexible load. Right: zoom 40–150 Hz.

Consistency between the axes. The fork mode seen by the azimuth encoder (55.8 Hz) and the elevation stator mass 1 (57.7 Hz) are within 2 Hz. This agrees with the hypothesis that the resonances near 56 Hz on both axes are the same vibration of the fork. The two axes are still fitted separately: the frequencies are not shared parameters.

6 Fit quality in the band 1–150 Hz

Table 6: Fit (NRMSE, %) on the measured points in the fit band and in its sub-bands, and rms errors over the band. "–": no points in the sub-band. H7: the configuration, not fitted.

axis	model	measurement	1–150	1–5	5–100	100–150	rms [dB]	rms [deg]
elevation	H7 + 2 masses	elevation, 1 % grid (1-600 Hz)	57.0	59.7	32.2	74.5	2.19	17.9
elevation	H7 + 2 masses	elevation, 1 % grid (30-150 Hz)	69.1	–	67.7	81.9	1.98	7.6
elevation	H7 + 2 masses	elevation, 16 screws, 4 A (30-300 Hz)	51.7	–	49.6	68.4	1.67	10.6
elevation	H7	elevation, 1 % grid (1-600 Hz)	37.9	30.1	24.8	−181.6	9.06	52.7
elevation	H7	elevation, 1 % grid (30-150 Hz)	−13.0	–	10.6	−167.1	15.40	86.9
elevation	H7	elevation, 16 screws, 4 A (30-300 Hz)	−11.7	–	8.3	−173.4	15.62	88.8
azimuth	flexible load	azimuth, 3 % grid (1-600 Hz)	69.6	68.2	62.0	6.7	2.14	8.6
azimuth	H7	azimuth, 3 % grid (1-600 Hz)	−21.1	−44.9	−16.0	−556.8	4.41	30.6

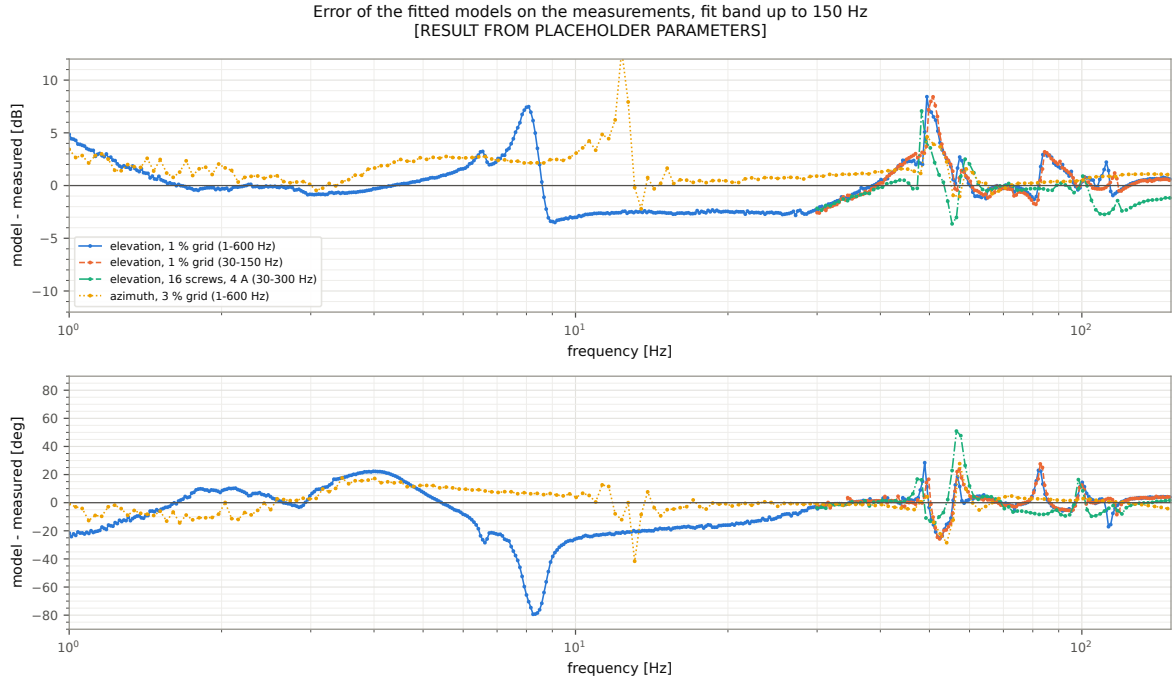


Figure 9: Error of the fitted models on each measurement in the fit band: magnitude [dB] and phase [deg], ε_i converted from Np and rad.

Where the models miss (Figures 5, 8 and 9).

- **Elevation, 7–10 Hz.** The measured antiresonance at 8.1 Hz and resonance at 9 Hz are not in the model: up to 8 dB and 80° of error. This is the price of removing the mass at 8.8 Hz, and it explains the 32 % of the 1–600 Hz measurement between 5 and 100 Hz.
- **Elevation, 40–55 Hz.** The measured small antiresonance at 48–51 Hz is not modelled, and the model lies 1–2 dB above the measurement before the peak. The mass at 53.4 Hz of the four-mass model covered this region.
- **Elevation, 56–59 Hz.** The measured double peak (56.3 and 58.5 Hz) is one peak at 57.6 Hz in the model.
- **Elevation, 84 Hz.** The small measured feature is not modelled.
- **Azimuth, 48–50 Hz.** The small measured pair of features is not modelled.

Table 7: Measured and modelled antiresonances and resonances in the fit band (peaks of at least 3 dB prominence).

curve	antiresonances [Hz]	resonances [Hz]
measured, elevation, 1 % grid (1-600 Hz)	8.08, 49.43, 83.76	56.25, 58.54, 99.19
measured, elevation, 1 % grid (30-150 Hz)	50.83, 84.44	56.71, 100.00
measured, elevation, 16 screws, 4 A (30-300 Hz)	48.25	56.54, 100.40
measured, azimuth, 3 % grid (1-600 Hz)	12.34, 64.57	55.70
model, elevation	–	57.55, 100.15
model, azimuth	12.88, 64.00	54.82

Two masses against four. Table 8 compares the reported model with the previous four-mass one (masses at 8.8, 53.4, 57.5, 100.2 Hz), fitted on the same data and band.

- The total elevation cost rises from 12.51 to 18.67, almost all of it on the 1–600 Hz measurement below 50 Hz.
- On the two measurements that start at 30 Hz, the two-mass model is slightly better. Without the low-frequency masses, the robust cost spends its effort on the fork modes.

Table 8: Elevation: fit (NRMSE, %) of the four-mass (4m) and of the two-mass (2m) models in the fit band and its sub-bands, and rms phase error [deg].

	1–150 Hz		1–5 Hz		5–100 Hz		100–150 Hz		rms [deg]	
measurement	4m	2m	4m	2m	4m	2m	4m	2m	4m	2m
elevation, 1 % grid (1-600 Hz)	69.0	57.0	61.0	59.7	66.0	32.2	71.5	74.5	11.5	17.9
elevation, 1 % grid (30-150 Hz)	65.9	69.1	–	–	64.2	67.7	81.8	81.9	6.9	7.6
elevation, 16 screws, 4 A (30-300 Hz)	49.4	51.7	–	–	46.8	49.6	73.6	68.4	11.6	10.6

7 Limitations and open points

- **Modal inertias.** Only $a_r = \kappa_r / J_r$ is identifiable. A physical modal inertia would need the motion of the stator measured independently, for example with an accelerometer on the fork.
- **Low frequency.** The elevation pair at 8–9 Hz is left out by choice. If the loop design needs it, the mass at about 9 Hz can be put back (three masses).
  - Its sharp antiresonance and damped resonance were reproduced only with an equivalent damping ratio of 0.17.
  - This suggests a friction-related, amplitude-dependent mechanism.
- **Telescope joints.** Neither k_b (elevation) nor k_3, c_3 (azimuth) are determined in the band. Both data sets only require the telescope to follow its support up to 150 Hz.
- **Dampers of the azimuth chain.** c_{LA} and c_2 are at their lower bound, and the azimuth delay is not determined.
- **Linear model.** The bearing constraint is a linear hysteretic equivalent fitted at the amplitude of these tests (3–4 A); it does not hold for other amplitudes.
- **Separate fits.** The two axes are fitted separately. A joint fit with shared fork modes, and the use of the coherence once it is exported, are the natural next steps.